FLCR-22 - Protein Descriptor And Fold-Facing Kernel

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper formalizes protein fold descriptors as an applied LCR-facing kernel. The operative object is fold descriptor kernel. The core result is that contact, homology, and winding descriptors can be staged as internal addressable features before biological validation. The paper also defines how this result routes forward: applied biology workbooks may consume descriptors only with PDB/parser and benchmark receipts. Its main residue is explicit: native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind.

Keywords

fold descriptor kernel; LCR; receipt; claim lane; normal form.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Protein Descriptor And Fold-Facing Kernel**. The paper's immediate contribution is: Defines protein descriptor routing and benchmark obligations.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Protein Descriptor And Fold-Facing Kernel.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines fold descriptor kernel as a first-class FLCR object.
- States the local result: contact, homology, and winding descriptors can be staged as internal addressable features before biological validation.
- Routes downstream use through claim lanes rather than inherited prose: applied biology workbooks may consume descriptors only with PDB/parser and benchmark receipts.
- Preserves the residue boundary: native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- `23_FoldForge_Protein_Folding.md`
- `virtuous/23_FoldForge_Protein_Folding_VIRTUOUS.md`
- `supplements/23_INTERNAL_REASONING_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Fold descriptor.** A protein-facing feature row derived from the applied forge reader.
- **Native-structure boundary.** The external validation gate for real protein structure claims.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

4. Formal Claims

Claim	Lane	Statement
Theorem 22.1	`normal_form_result`	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation
Proposition 22.2	`normal_form_result`	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane.
Protocol 22.3	`falsifier_or_open_obligation`	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 22.1	formal construction	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 22.2	formal construction	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 22.3	obligation/falsifier	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	`FLCR-22` defines or uses **Protein Descriptor And Fold-Facing Kernel** as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-22.
2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.
3. State the strongest admissible claim in its current lane.
4. Export only the lane-safe result to downstream papers and preserve the residue.

Proof Sketch

The proof strategy for FLCR-22 is a typed construction argument. The paper names fold descriptor as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is `Protein Descriptor And Fold-Facing Kernel`. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Legacy paper body	Primary formal and source-integration material assigned by the series map.
Internal and pyramid supplements	Cross-paper reasoning windows, deferred back-application slots, and route constraints.
NSIT tool closure matrix	Existing verifier/tool families that can close internal or batch claims.
CAM/crystal and standards-window evidence	Projected memory, source routing, standards reports, and window/node databases.

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GAUSSHAUS-FORGE-TOPOLOGY	The forge topology describes a typed family of expression surfaces: candidate generation, lattice resolution, CAD/geometry boundary, manufacturing process bridge, SplatForge readout, and simulation/test comparison.	When reapplied, the topology becomes the general expression grammar for the system: every applied paper can say which forge surface is acting, what it reads, what it writes, which validator promotes it, and which residue remains.	Lattice Forge and the forges stop reading as separate tools; they become the way CAM is expressed into domain action.; Materials, proteins, games, plasma, computation, and publication tooling can share a typed read/write/receipt surface.; Physical-realization boundaries become clearer because CAD, process, simulation, printed artifact, and measurement are not collapsed into one result.	The topology does not manufacture external test data.; Every physical claim still requires machine, process, simulation, and metrology receipts where relevant.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GAUSSHAUS-FORGE-TOPOLOGY	`CAM_crystal_reapplication_result`	Adds a typed forge family model: MetaForge proposes candidates, LatticeForge resolves local architecture, CADForge creates inspectable fabrication geometry, the Process Bridge gates manufacturing claims, SplatForge reads the state, and simulation/test bridges compare prediction to observation. Evidence facts: forge surfaces: MetaForge, LatticeForge, CADForge, Process Bridge, SplatForge/GaussHaus Field, simulation/test bridges; typed L/C/R surface: data in/out, control plane, outward receipt surface.	This closes expression architecture for applied materials and forge papers; each physical realization still requires machine, process, simulation, and metrology receipts.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	`receipt_bound_internal_result`	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.507121.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
carrier enumeration	`normal_form_result`	The LCR carrier action is closed as an addressable finite-state surface for its declared domain.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-3 tower	`receipt_bound_internal_result`	The decade tower is resolved with avg TarPit mass=0.509294 and mass sum=40.743500.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-02 cross-lift comparison	`cam_crystal_reapplication_result`	Family tower binds FLCR-02, FLCR-12, FLCR-22, FLCR-32; avg TarPit mass=0.511700; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
protein descriptor kernel	`normal_form_result`	Protein descriptor and fold-facing state are closed as descriptor surfaces.	Biophysical folding accuracy requires benchmark data.

Claim Posture After This Pass

`FLCR-22` should now be read as a resolved-state contribution for `Protein Descriptor And Fold-Facing Kernel` at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- `FLCR-11` through `FLCR-18` for window, receipt, admission, CA, algebraic,

curvature, residue, and exceptional machinery.

- `FLCR-28` through `FLCR-30` for CAM/crystal, corpus ribbon, and universal

normal-form intake.

- `FLCR-31` through `FLCR-40` only after the LCR-native result has been cited

and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind
- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 23/fold_action.

Original slot	Family	Lift	Role	Proof form
23	fold_action	2	order-3 slot-3: fold the initial action into a higher-dimensional representation	fold map, coordinate atlas, and representation transport

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 22 and reopened at original slot 23 (FoldForge Protein Folding).

It must produce: formal result of original slot 23 plus the same-family lift path toward slot 33.

Semantic transition: fold the local state into a higher representation while preserving addressability.

Accepted formalism targets to bind in the journal rewrite:

- representation theory
- group actions and equivariance
- tensor or coordinate atlases
- transport maps between equivalent descriptions

Slot	Family	Lift	Produced proof form
23	fold_action	2	fold map, coordinate atlas, and representation transport

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to strengthen representation transport: every face, gauge, atlas, or fold label must point to a preserved source object.

Primary Evidence Inputs

Source	Placement reason
`23_FoldForge_Protein_Folding.md`	primary assigned source for the paper's formal spine
`supplements/23_INTERNAL_REASONING_SUPPLEMENT.md`	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations
`virtuous\23_FoldForge_Protein_Folding_VIRTUOUS.md`	prior enriched paper body; should be mined for mature claims and proof ordering

Secondary And Orbital Evidence Inputs

Source	Placement reason
`supplements/METAFORGE_MATERIALS_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/RECEIPT_VERIFIER_CATALOG.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/APPLIED_FORGES_WORKBOOK.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_PYRAMID_INDEX.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`CAM_CLAIM_100_SCORE_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_TOOL_CLOSURE_MATRIX.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_REASONED_CLOSURE_PASS.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`ORBITAL_DATA_CROSS_POPULATION_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`AGENT_CRYSTAL_WORK_HARVEST.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.

- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

23	74	applied partial closure	FoldForge contact/homology/winding framing, parser route, applied forge validation lane	PDB/native structure parser and fold-rate validation

Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Applied Forge Internal/External Validation Split	`receipt_bound_internal_result`	Promote internal forge reader, descriptor, candidate-generation, and finite validation kernels when validators pass.	Attach forge tool path, input object, output descriptor/candidate, validator result, and explicit external-validation boundary.	Wet-lab, fabrication, biological, gameplay-global, or real-world optimization claims remain external-validation needs.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Applied Forge Internal External Split	This paper may claim an internal forge reader/generator/descriptor kernel where the tool produces replayable internal outputs.	Forge validators, applied-forge workbooks, material/protein/game receipts, CAM addresses, and source-routing rows.	Promote internal representation/generation kernels when validators pass, while routing real-world validation separately.	Fabrication, wet-lab, biological, gameplay-global, manufacturing, or measured optimization claims remain external-validation needs.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
FORMAL: Paper 23 — C-Form: FoldForge Protein Folding Gluon	89	**C = the protein fold Gluon** — the contact-map/topo Gluon that transports protein chain fold hypotheses through contact-map and topology receipts. In the lattice_forge substrate, C is realized as the **fold Gluon** ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 22 — C-Form: MetaForge Applied Materials Gluon	85	**C = the material Gluon** — the applied materials Gluon that transports the morphic Gluon (Paper 21) into physical material candidates. In the lattice_forge substrate, C is realized as the **material Gluon** that: - ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
SUMMARY-III-Computational-Substrates: Summary Paper III — Computational Substrates: The Gluon as Fold, Knight, Traversal, Pinch, Delay, Game, Monster	83	This paper presents the **computational substrate applications** of the Gluon formalism. We do not use the word "Gluon" because the fit is good. We use it because the substrate **IS** SU(3), and the computational subs...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 25 — C-Form: Energetic Traversal Maps Gluon	81	**C = the traversal energy Gluon** — the energy/ledger Gluon that adds energy and traversal costs to cross-language, figure, material, and fold transformations. In the lattice_forge substrate, C is realized as the **t...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-22.75: Paper 22.75 - MetaForge Next-State Bridge	81	Paper 22 exports a candidate-ledger pattern to the next applied papers. The pattern is: choose a domain inventory, score admissible partners or transforms, run the finite candidate transport, preserve failures as obli...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	18 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	`FLCR-22`
Legacy source slot(s)	`23`
Ribbon role	order-3 slot-3: fold the initial action into a higher-dimensional representation
Proof form	fold map, coordinate atlas, and representation transport
Received state	state emitted by prior slot 22 and reopened at original slot 23 (FoldForge Protein Folding)
Produced state	formal result of original slot 23 plus the same-family lift path toward slot 33

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 22.1	`normal_form_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation
Proposition 22.2	`normal_form_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane.

Claim	Lane	Evidence to attach	Boundary
Protocol 22.3	`falsifier_or_open_obligation`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind

Figure Plan

Figure	Purpose	Caption
FLCR-22-F1	Representation fold atlas	Schematic showing how `FLCR-22` turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-22-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-22-F3	Same-family lift context	Diagram placing this paper in the original `00-79` ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-22-F1: Representation fold atlas

```
received state
  -> Representation fold atlas
  -> formal claim lane
  -> evidence object / receipt / citation
  -> produced state
  -> remaining residue
```

Table Plan

Table	Purpose
FLCR-22-T1	Face/representation correspondence table
FLCR-22-T2	Claim-lane/evidence-boundary table
FLCR-22-T3	Archive evidence card and source-backed upgrade table
FLCR-22-T4	Workbook replay and falsifier table

Worked Example

Map one source object across two labels/faces while preserving its address. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
atlas	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
face	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
fold	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
representation transport	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-22-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

`FLCR-22` (`Protein Descriptor And Fold-Facing Kernel`) occupies the `fold_action` role at lift depth 2. The paper receives state emitted by prior slot 22 and reopened at original slot 23 (FoldForge Protein Folding). It produces formal result of original slot 23 plus the same-family lift path toward slot 33. The state transition is: fold the local state into a higher representation while preserving addressability.

The paper-local proof form is:

```
fold map, coordinate atlas, and representation transport
```

The integration result is a representation-transport chart with named invariants and residues. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
representation theory	Formal carrier for the paper-local result.
group actions and equivariance	Formal carrier for the paper-local result.
tensor or coordinate atlases	Formal carrier for the paper-local result.
transport maps between equivalent descriptions	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 22.1	`normal_form_result`	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation
Proposition 22.2	`normal_form_result`	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane.
Protocol 22.3	`falsifier_or_open_obligation`	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	`23_FoldForge_Protein_Folding.md`, `virtuous/23_FoldForge_Protein_Folding_VIRTUOUS.md`, `supplements/23_INTERNAL_REASONING_SUPPLEMENT.md`	routed evidence
Evidence inputs	`CAM_CLAIM_100_SCORE_AUDIT.md`, `NSIT_TOOL_CLOSURE_MATRIX.md`, `NSIT_REASONED_CLOSURE_PASS.md`, `PAPER_REWORKS_CRYSTAL_PROJECTION.json`	routed evidence
CAM/crystal sources	`PAPER_REWORKS_CRYSTAL_PROJECTION.json`, `AGENT_CRYSTAL_WORK_HARVEST.json`, `runtime standards artifact`, `notebook-derived source bundle`	routed evidence
Standards-window inputs	`window_summary.json`, `node DBs`, `window DBs`	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting

the operation;

4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-22-F1: Representation fold atlas	Visualizes the proof object, routing, or boundary.
FLCR-22-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-22-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-22-T1: Face/representation correspondence table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-22-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.

Table	Role
FLCR-22-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-22-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.